CODEORBIT TECH

EXPLORATORY DATA ANALYSIS (EDA)

INTERNSHIP TASK 2

Visual exploration of passenger characteristics and survival patterns

Dataset: Titanic passenger dataset (891 records, 12 columns)

Tools: Python, Jupyter Notebook, Pandas, NumPy, Matplotlib and Seaborn

Prepared for internship task submission

1. Executive Summary

This report summarizes exploratory data analysis of the Titanic dataset using Pandas, Matplotlib and Seaborn. The notebook first inspected the original data and then loaded the cleaned output from Task 1 for the main visual analysis. The analysis includes univariate distributions, comparisons between variables, a correlation heatmap and box plots for potential outlier inspection.

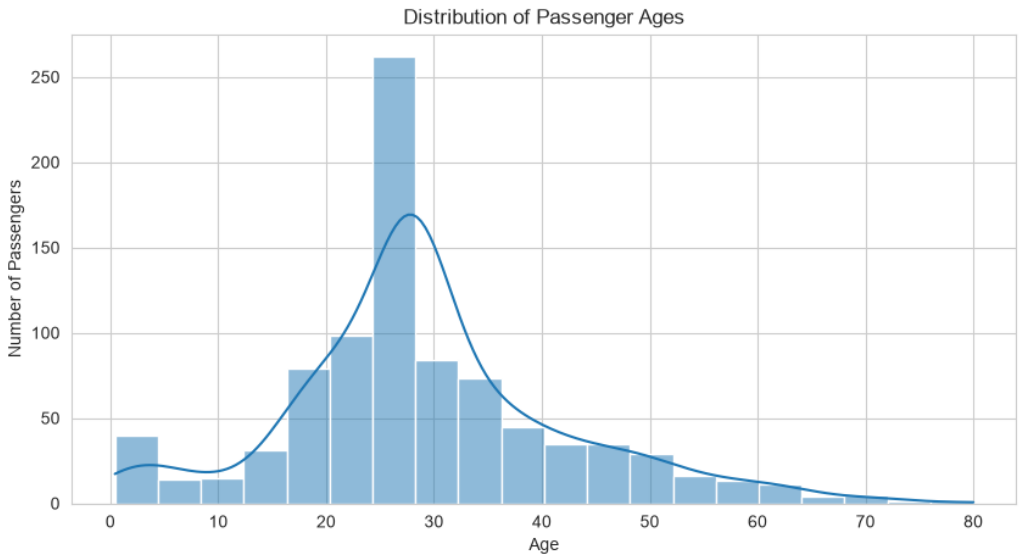
2. Dataset and Analysis Basis

Item	Description
Original dataset	891 records and 12 columns
Main EDA input	Titanic_Cleaned.csv produced in Task 1
Libraries	Pandas, NumPy, Matplotlib, Seaborn
Analysis types	Univariate, bivariate and numerical correlation analysis

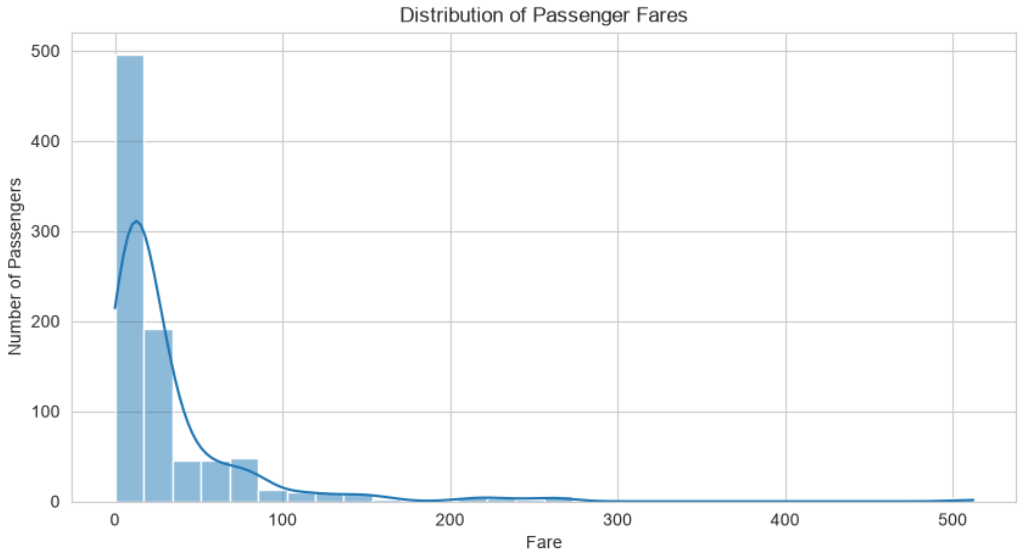
The original data contained missing values in Age (177), Cabin (687) and Embarked (2). For the main EDA, the notebook loaded the cleaned dataset: Age was median-imputed, Embarked was mode-imputed and Cabin missing entries were labelled Unknown. This is important because the EDA charts are based on the cleaned values, not on the original missing-value representation.

3. Univariate Analysis

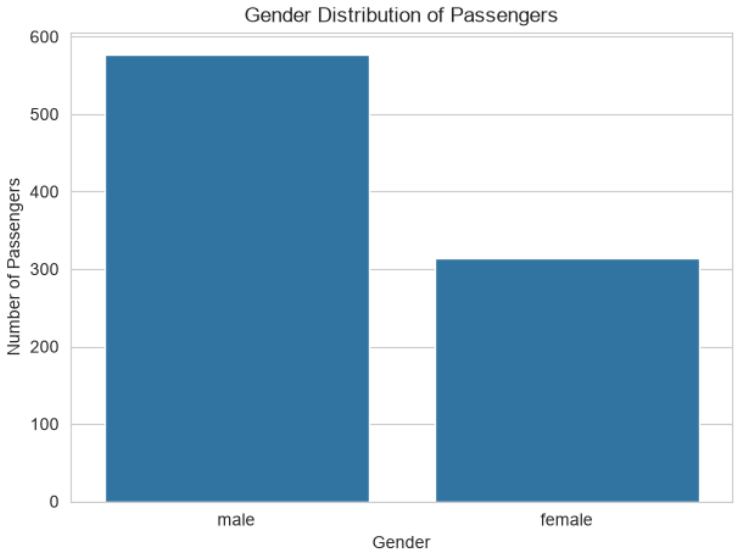
The notebook examined the distribution of passenger Age and Fare using histograms, and summarized counts for Sex, Pclass and Survived using bar charts.



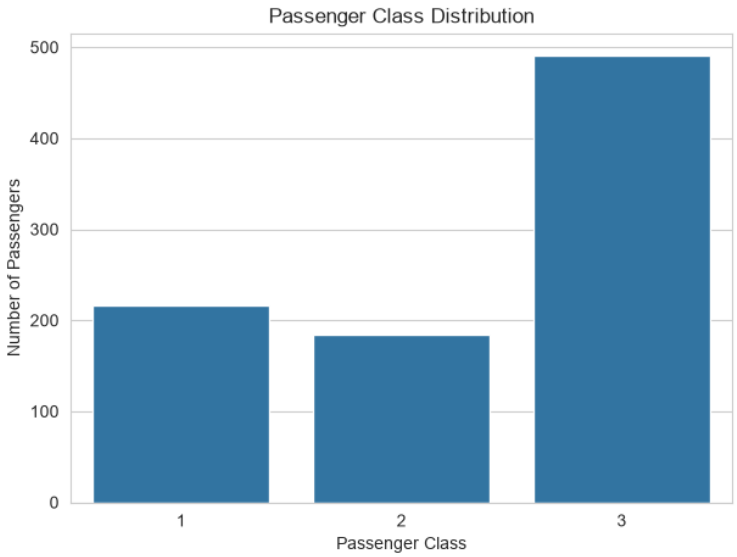
Passenger age distribution



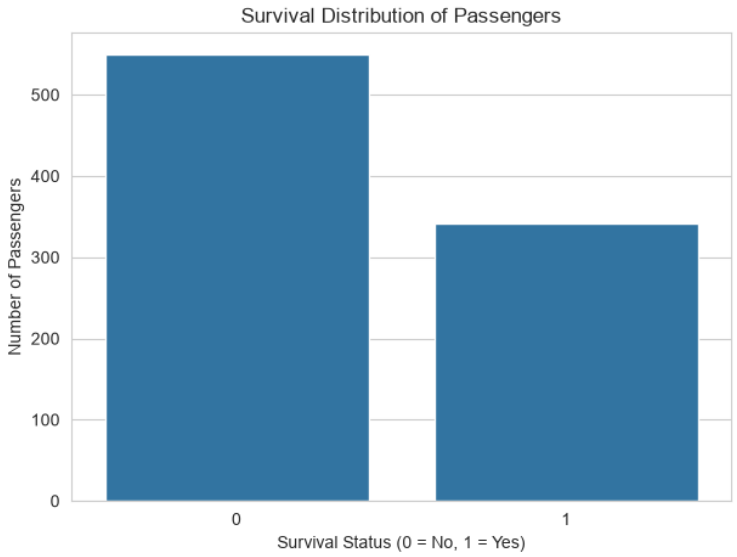
Passenger fare distribution



Passenger gender counts



Passenger class counts

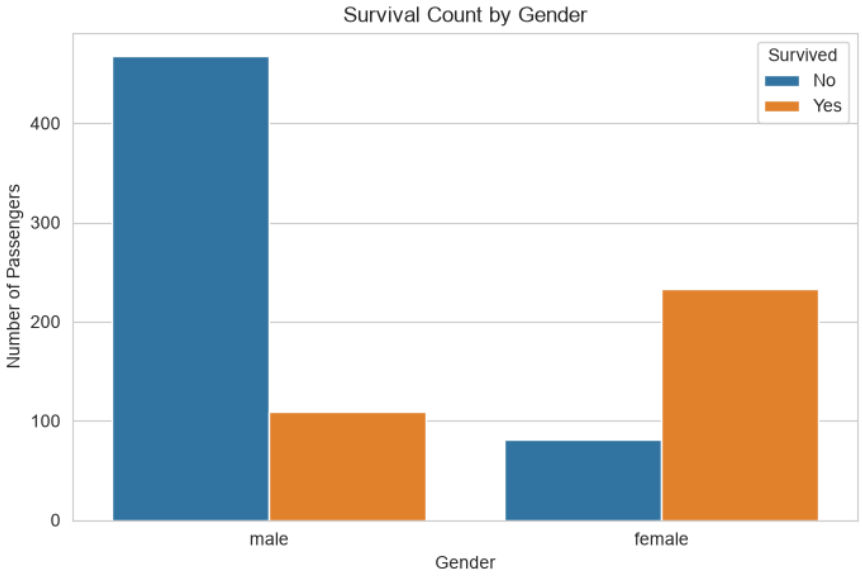


Survival counts

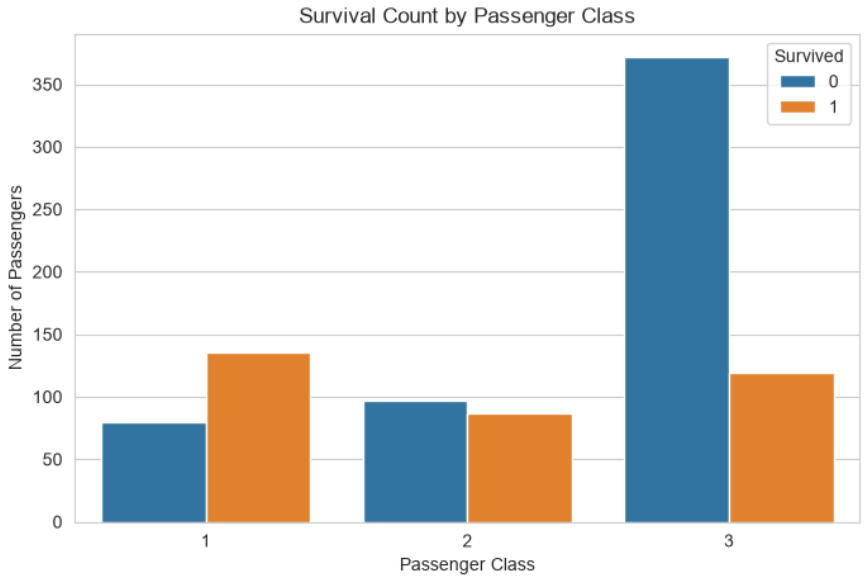
Observed counts: the dataset contains 577 male passengers and 314 female passengers. There are 549 non-survivors and 342 survivors. These counts describe the sample composition; survival rates by group are examined separately below.

4. Bivariate Analysis

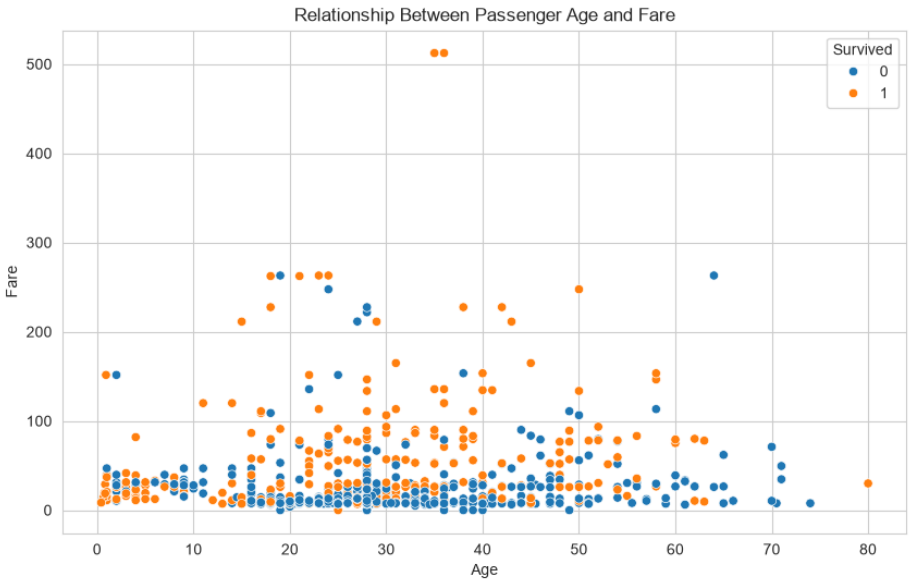
The notebook compared survival counts by gender and passenger class, plotted Age against Fare with points coloured by survival status, and compared Age distributions for survivors and non-survivors.



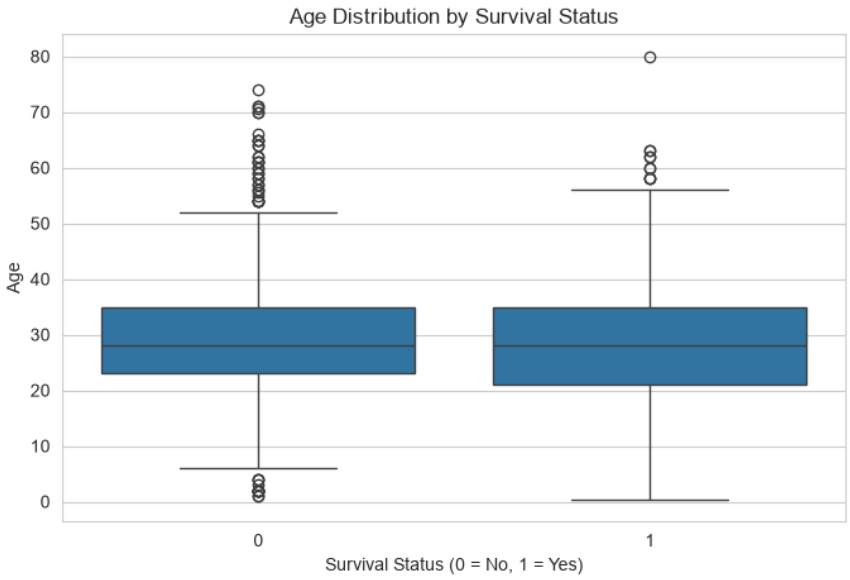
Survival counts by gender



Survival counts by passenger class



Passenger age versus fare



Age distribution by survival status

Survival by gender

Gender	Did not survive	Survived
Female	25.80%	74.20%
Male	81.11%	18.89%

In this dataset, 74.20% of female passengers survived compared with 18.89% of male passengers. These are within-gender percentages, calculated by dividing each gender's survival counts by its total passenger count.

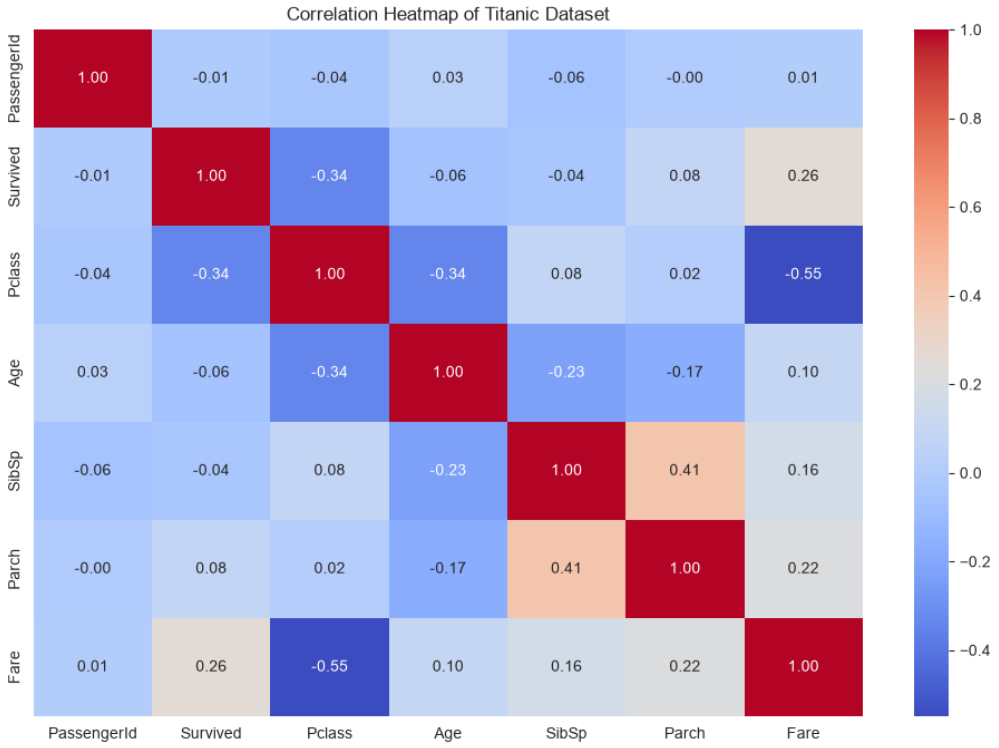
Survival by passenger class

Passenger class	Did not survive	Survived
1st class	37.04%	62.96%
2nd class	52.72%	47.28%
3rd class	75.76%	24.24%

The first-class group had the highest survival percentage (62.96%), followed by second class (47.28%) and third class (24.24%). These results describe an association in this dataset and should not be interpreted as proof that passenger class alone caused survival differences.

5. Multivariate / Correlation Analysis

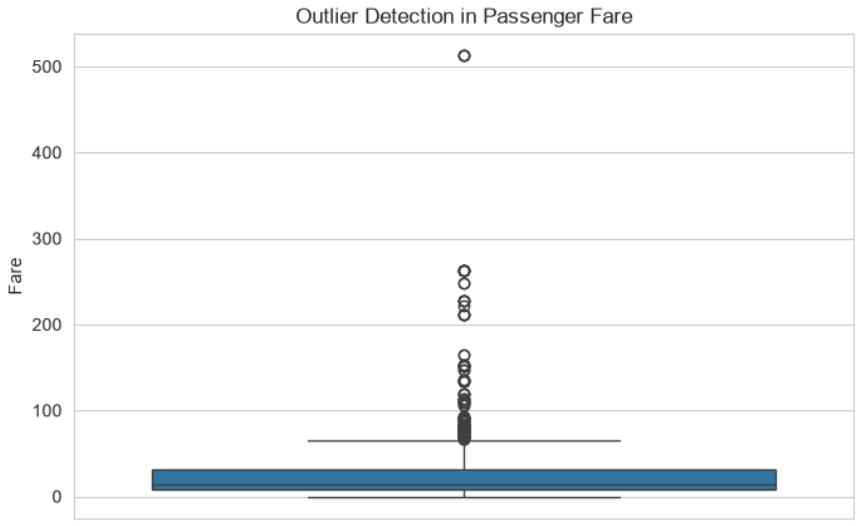
The notebook calculated a correlation matrix for numerical columns and visualized it as a heatmap. Correlation indicates the direction and strength of linear association between numerical variables; it does not establish causation. PassengerId is an identifier and should not be interpreted as a meaningful measured feature.



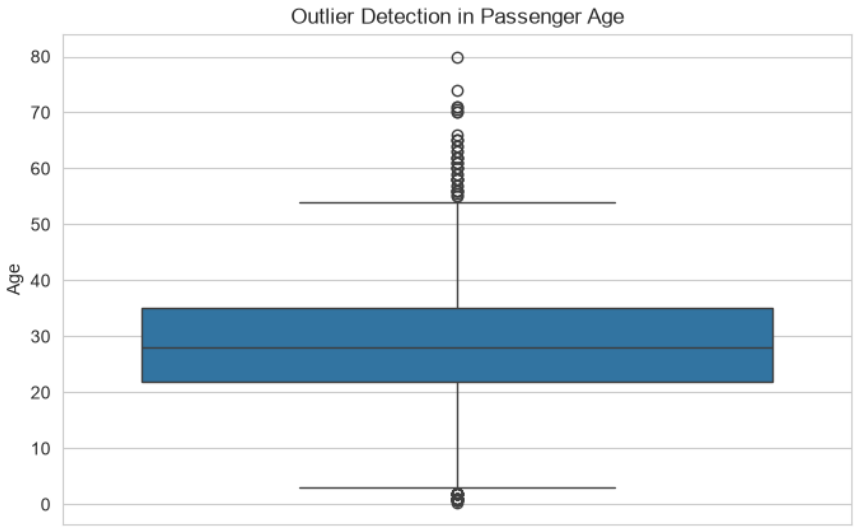
Correlation heatmap

6. Outlier Inspection

Box plots were generated for Fare and Age to visually inspect potential outliers. The interquartile range (IQR) rule was also calculated for this report to quantify observations beyond 1.5 × IQR from the quartiles. These points are flagged for review, not automatically treated as errors.



Fare outlier box plot



Age outlier box plot

Variable	Q1	Q3	IQR	IQR outlier count	Observed range
Age	22.00	35.00	13.00	66	0.42 – 80.00
Fare	7.91	31.00	23.09	116	0.00 – 512.33

Using the 1.5 × IQR rule on the cleaned dataset, 66 Age observations and 116 Fare observations are flagged. The Fare range extends to 512.33, while the Age range extends from 0.42 to 80 years. These values should be examined in context; unusual values can be legitimate passenger records.

7. Key Findings

- The dataset contains 891 passengers; 342 (38.38%) survived and 549 (61.62%) did not survive.
- Female passengers had a higher observed survival percentage (74.20%) than male passengers (18.89%).
- First-class passengers had the highest observed survival percentage (62.96%), while third-class passengers had the lowest (24.24%).
- The median Age after cleaning is 28 years and the mean Age is approximately 29.36 years.
- The Fare distribution includes high-value observations; the IQR method flags 116 records for review. Age has 66 IQR-flagged observations.

8. Conclusion

The EDA identified clear differences in observed survival percentages across gender and passenger class, described the distributions of Age and Fare, and highlighted potential outliers for further review. The visualizations and summary statistics provide a descriptive understanding of the Titanic dataset. The analysis is observational and does not establish causal explanations for the patterns.

Report preparation note: The uploaded EDA notebook contains executed charts and computed survival percentages, but does not contain written chart-by-chart interpretations. This report adds concise interpretations calculated from the supplied Titanic CSV and the cleaning procedure documented in Task 1.